

 **cl-nagoya/ruri-v3-310m** 

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Sentence Similarity



Safetensors



cl-nagoya/ruri-v3-dataset-ft



Japanese

modernbert

feature-extraction

 arxiv:2409.07737



License: apache-2.0



Model card



Files



xet



Community

Downloads last month

100,479



 Safetensors 

Model size

315M params

Tensor type

F32

 Files info

 Inference Providers NEW

 Sentence Similarity

This model isn't deployed by any Inference Provider.



1

Ask for provider support

 Model tree for cl-nagoya/ruri-v3-310m

Base model

[sbintuitions/modernbert-ja-310m](#)



Finetuned

[cl-nagoya/ruri-v3-pt-310m](#)



Finetuned (4)

this model

Quantizations

3 models

 Dataset used to train cl-nagoya/ruri-v3-310m

 Space using cl-nagoya/ruri-v3-310m 1

 Collection including cl-nagoya/ruri-v3-310m

Ruri v3

 Collection

Japanese General Text Embeddings with ... • 14 items • Updated May 10 •  2

🔗 Ruri: Japanese General Text Embeddings

Ruri v3 is a general-purpose Japanese text embedding model built on top of **ModernBERT-Ja**. Ruri v3 offers several key technical advantages:

- **State-of-the-art performance** for Japanese text embedding tasks.
- **Supports sequence lengths up to 8192 tokens**
 - Previous versions of Ruri (v1, v2) were limited to 512.
- **Expanded vocabulary of 100K tokens**, compared to 32K in v1 and v2
 - The larger vocabulary make input sequences shorter, improving efficiency.
- **Integrated FlashAttention**, following ModernBERT's architecture
 - Enables faster inference and fine-tuning.
- **Tokenizer based solely on SentencePiece**
 - Unlike previous versions, which relied on Japanese-specific BERT tokenizers and required pre-tokenized input, Ruri v3 performs tokenization with SentencePiece only—no external word segmentation tool is required.

🔗 Model Series

We provide Ruri-v3 in several model sizes. Below is a summary of each model.

ID	#Param.	#Param. w/o Emb.	Dim.	#Layers	Avg. JMTEB
cl-nagoya/ruri-v3-30m	37M	10M	256	10	74.51
cl-nagoya/ruri-v3-70m	70M	31M	384	13	75.48
cl-nagoya/ruri-v3-130m	132M	80M	512	19	76.55
cl-nagoya/ruri-v3-310m	315M	236M	768	25	77.24

🔗 Usage

You can use our models directly with the transformers library v4.48.0 or higher:

```
pip install -U "transformers>=4.48.0" sentence-transformers
```

Additionally, if your GPUs support Flash Attention 2, we recommend using our models with Flash Attention 2.

```
pip install flash-attn --no-build-isolation
```

Then you can load this model and run inference.

```
import torch
import torch.nn.functional as F
from sentence_transformers import SentenceTransformer

# Download from the 🤗 Hub
device = "cuda" if torch.cuda.is_available() else "cpu"
model = SentenceTransformer("cl-nagoya/ruri-v3-310m", device=device)

# Ruri v3 employs a 1+3 prefix scheme to distinguish between different types of tokens.
# "" (empty string) is used for encoding semantic meaning.
# "トピック: " is used for classification, clustering, and encoding topical information.
# "検索クエリ: " is used for queries in retrieval tasks.
# "検索文書: " is used for documents to be retrieved.
sentences = [
    "川|べりでサーフボードを持った人たちがいます",
    "サーファーたちが川|べりに立っています",
    "トピック: 瑠璃色のサーファー",
    "検索クエリ: 瑠璃色はどんな色?",
    "検索文書: 瑠璃色（るりいろ）は、紫みを帯びた濃い青。名は、半貴石の瑠璃（ラピスラズリ、英：1
]

embeddings = model.encode(sentences, convert_to_tensor=True)
print(embeddings.size())
```

```
# [5, 768]
```

```
similarities = F.cosine_similarity(embeddings.unsqueeze(0), embeddings.unsqueeze(1))
print(similarities)
# [[1.0000, 0.9603, 0.8157, 0.7074, 0.6916],
#  [0.9603, 1.0000, 0.8192, 0.7014, 0.6819],
#  [0.8157, 0.8192, 1.0000, 0.8701, 0.8470],
#  [0.7074, 0.7014, 0.8701, 1.0000, 0.9746],
#  [0.6916, 0.6819, 0.8470, 0.9746, 1.0000]]
```

[Benchmarks](#)

[JMTEB](#)

Evaluated with [JMTEB](#).

Model	#Param.	Avg.	Retrieval	STS	Classification	Reranking	Clustering	Pairwise
Ruri-v3-30m	37M	74.51	78.08	82.48	74.80	93.00	52.12	
Ruri-v3-70m	70M	75.48	79.96	79.82	76.97	93.27	52.70	
Ruri-v3-130m	132M	76.55	81.89	79.25	77.16	93.31	55.36	
Ruri-v3-310m (this model)	315M	77.24	81.89	81.22	78.66	93.43	55.69	
sbintuitions/sarashina-embedding-v1-1b	1.22B	75.50	77.61	82.71	78.37	93.74	53.86	
PLaMo-Embedding-1B	1.05B	76.10	79.94	83.14	77.20	93.57	53.47	
OpenAI/text-embedding-ada-002	-	69.48	64.38	79.02	69.75	93.04	48.30	
OpenAI/text-embedding-3-small	-	70.86	66.39	79.46	73.06	92.92	51.06	

Model	#Param.	Avg.	Retrieval	STS	Classification	Reranking	Clustering	Pai
OpenAI/text-embedding-3-large	-	73.97	74.48	82.52	77.58	93.58	53.32	
pkshatech/GLuCoSE-base-ja	133M	70.44	59.02	78.71	76.82	91.90	49.78	
pkshatech/GLuCoSE-base-ja-v2	133M	72.23	73.36	82.96	74.21	93.01	48.65	
retrieva-jp/amber-base	130M	72.12	73.40	77.81	76.14	93.27	48.05	
retrieva-jp/amber-large	315M	73.22	75.40	79.32	77.14	93.54	48.73	
sentence-transformers/LaBSE	472M	64.70	40.12	76.56	72.66	91.63	44.88	
intfloat/multilingual-e5-small	118M	69.52	67.27	80.07	67.62	93.03	46.91	
intfloat/multilingual-e5-base	278M	70.12	68.21	79.84	69.30	92.85	48.26	
intfloat/multilingual-e5-large	560M	71.65	70.98	79.70	72.89	92.96	51.24	
Ruri-Small	68M	71.53	69.41	82.79	76.22	93.00	51.19	
Ruri-Small v2	68M	73.30	73.94	82.91	76.17	93.20	51.58	
Ruri-Base	111M	71.91	69.82	82.87	75.58	92.91	54.16	
Ruri-Base v2	111M	72.48	72.33	83.03	75.34	93.17	51.38	
Ruri-Large	337M	73.31	73.02	83.13	77.43	92.99	51.82	
Ruri-Large v2	337M	74.55	76.34	83.17	77.18	93.21	52.14	

🔗 Model Details

🔗 Model Description

- **Model Type:** Sentence Transformer
- **Base model:** [cl-nagoya/ruri-v3-pt-310m](#)
- **Maximum Sequence Length:** 8192 tokens
- **Output Dimensionality:** 768
- **Similarity Function:** Cosine Similarity
- **Language:** Japanese
- **License:** Apache 2.0
- **Paper:** <https://arxiv.org/abs/2409.07737>

🔗 Full Model Architecture

```
SentenceTransformer(  
  (0): Transformer({'max_seq_length': 8192, 'do_lower_case': False}) with Transfo  
  (1): Pooling({'word_embedding_dimension': 768, 'pooling_mode_cls_token': False,  
  )
```

🔗 Citation

```
@misc{  
  Ruri,  
  title={{Ruri: Japanese General Text Embeddings}},  
  author={Hayato Tsukagoshi and Ryohei Sasano},  
  year={2024},  
  eprint={2409.07737},  
  archivePrefix={arXiv},  
  primaryClass={cs.CL},  
  url={https://arxiv.org/abs/2409.07737},  
}
```

License

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